# GARRY BAJWA

## AUTOMATION MANAGER | INDUSTRIAL CONTROLS | EQUIPMENT RELIABILITY | MANUFACTURING SYSTEMS

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## PROFESSIONAL SUMMARY

Automation, maintenance and manufacturing systems leader with 7+ years of experience across industrial equipment, electronics engineering, plant operations, controls troubleshooting and technical team leadership. Hands-on experience supporting PLC-controlled production machinery and diagnosing faults across control logic, I/O, sensors, actuators, motors, drives, industrial PCs, networks and electromechanical systems. Working knowledge of Ladder Logic, Structured Text concepts, PLC sequence troubleshooting and Siemens S7-1200/LOGO! platforms, backed by strong root-cause analysis, preventive/corrective maintenance, barcode/scanner systems, KPI reporting, technical documentation and continuous improvement.

## CORE TECHNICAL PROFILE

- PLC / controls: Troubleshooting PLC-controlled manufacturing equipment; reading machine sequences, interlocks and I/O states; Ladder Logic and Structured Text concepts; Siemens S7-1200 and LOGO! hands-on lab work; HMI, sensors, actuators and industrial control devices.
- Motor / machine systems: Motors, VFDs/drives, starters/control circuits, CNC and motion-control equipment, embedded controllers, industrial PCs, electrical/electronic diagnostics and cross-domain electromechanical troubleshooting.
- Reliability / maintenance: Preventive and corrective maintenance, breakdown response, work prioritization, CMMS/work-order workflow fundamentals, root-cause analysis, recurring-failure elimination, SOPs, technical documentation and equipment-upgrade coordination.
- Operations / systems integration: Barcode scanners, traceability, production data, KPI dashboards, industrial networking, Python-based workflow automation, vendor/contractor coordination, technical mentoring and cross-functional problem solving.

## PROFESSIONAL EXPERIENCE

**Systems, Automation & Maintenance Manager** **Jan 2024 - Sep 2026**

*Van Gogh Designs — Surrey, BC*

- Lead automation, equipment reliability, maintenance and manufacturing-systems initiatives across a multi-department production environment, working with production, scheduling, quality, supervisors, vendors, contractors and management.
- Support and troubleshoot PLC-controlled CNC and production machinery, tracing faults through machine sequences, interlocks, I/O, sensors, actuators, motors, VFD/drives, electrical controls, industrial PCs, networks and mechanical systems to restore safe operation.
- Apply PLC and Ladder Logic fundamentals during machine troubleshooting; use control-sequence and I/O understanding to distinguish programming/control faults from sensor, drive, electrical, network or mechanical failures, while continuing hands-on development with Siemens S7-1200 and LOGO! platforms.
- Lead root-cause investigations for recurring equipment and process failures, document failure modes and corrective actions, and implement reliability improvements intended to prevent repeat breakdowns and protect production uptime.
- Coordinate preventive and corrective maintenance priorities, breakdown response, contractor/vendor work and equipment upgrades; define repair scope, support technical decisions, verify completed work and communicate operational risks and workarounds.
- Develop and maintain manufacturing automation tools that integrate barcode scanners, workstations, industrial computers and production data for traceability, live status visibility, KPI reporting and workflow control.
- Improve shop-floor information flow through production tracking, automated reporting and exception visibility, reducing reliance on manual status checks and helping supervisors identify bottlenecks and pending work faster.
- Create troubleshooting guidance, SOPs and user documentation and train supervisors/operators on equipment use, escalation, process discipline and newly deployed manufacturing systems.

## SELECTED AUTOMATION & RELIABILITY CONTRIBUTIONS

- Implemented barcode/scanner-based production tracking and traceability tools that connect shop-floor activity with live operational status and reporting, improving visibility of pending work and production flow.
- Built automated KPI and exception-reporting workflows to surface bottlenecks, recurring issues and department performance without relying solely on manual data collection or status meetings.
- Supported maintenance and technical ownership across CNC equipment, production machinery, networking and connected shop-floor systems, coordinating internal troubleshooting with external vendor support when required.
- Combine controls, electronics, software and mechanical troubleshooting to isolate faults across complete machine systems rather than treating PLC, electrical, network and mechanical issues as separate problems.

## PROFESSIONAL EXPERIENCE - CONTINUED

**Contract Electronics Engineer / Engineering Team Lead** **Jun 2023 - Dec 2023**

*Tinari Energy Management Services Inc. — Canada*

- Helped lead development and integration of a large-scale cable-driven 3D concrete printing system using a four-tower architecture and multi-axis motion platform, combining motors, drives, sensors, controllers, industrial electronics, embedded interfaces, software/firmware and mechanical subsystems.
- Supported controls architecture, electrical/wiring integration, system assembly, bring-up, commissioning and functional testing across interconnected motion-control, electrical, software and mechanical systems.
- Commissioned and troubleshot motors, drives, sensors, controllers and embedded interfaces; investigated failures across subsystem boundaries and supported root-cause testing, corrective actions and design iterations.
- Worked with Trio 8-axis motion-control hardware, Galil/linear-actuator subsystems and multi-motor systems, supporting coordinated machine behaviour and interpreting technical requirements, wiring information and functional documentation during integration and troubleshooting.
- Supported HMI/IoT interfaces and collaborated with engineering team members to resolve technical issues, communicate implementation risks and recommendations, and move prototype subsystems toward stable operation.

**Production Supervisor / Technical Manufacturing Support** **Oct 2021 - Nov 2022**

*Signhub Canada — Surrey, BC*

- Supervised custom manufacturing workflows, employees and daily priorities across signage production, aluminum fabrication, welding and installation while maintaining safety, quality and delivery requirements.
- Supported CNC router, laser, printer/plotter and CAD/CAM processes; diagnosed equipment/process issues and translated design requirements into practical, repeatable manufacturing steps.
- Coordinated materials, production sequencing, technical problem solving and shop-floor execution in a deadline-driven manufacturing environment.

**SMT Operator, Part-Time** **Feb 2021 - Aug 2021**

*Motorola Solutions — Canada*

- Operated Juki pick-and-place equipment and supported PCB assembly, JTAG-based testing, quality checks and electronics manufacturing in a controlled production environment.

**Electronics / Electrical Engineer** **Feb 2019 - Nov 2020**

*Bajwa Smart Tech*

- Developed and troubleshot embedded electronic and control systems using C/C++, microcontrollers, sensors, motors, actuators and related interface electronics.
- Worked with circuit diagrams, PCB design, firmware, electrical/electronic troubleshooting and CNC-related control hardware, building a strong foundation in control-system fault isolation.
- Performed prototype assembly, testing, debugging and technical documentation across hardware and embedded-software components.

## TECHNICAL SKILLS

**Controls & Automation:** PLC-controlled equipment troubleshooting; Ladder Logic; Structured Text concepts; Siemens S7-1200; Siemens LOGO!; HMI; Trio Motion Controller; Fanuc troubleshooting; sensors; actuators; motors; VFD/drives; CNC and motion-control systems.

**Maintenance & Industrial:** PM/CM planning; breakdown troubleshooting; CMMS/work-order workflow fundamentals; root-cause analysis; equipment reliability; electrical/electronic diagnostics; industrial PCs; networking; compressors; dust collection; production machinery; technical documentation.

**Software & Manufacturing Systems:** Python; C/C++; Embedded C; Flask; pandas; OpenPyXL; MQTT; Raspberry Pi; Arduino/ESP; Linux/Windows; barcode traceability; production tracking; KPI dashboards; workflow automation.

**Engineering & Improvement:** Functional testing; commissioning; failure investigation; SOPs; Lean/Kaizen; CAD/CAM; Onshape; DraftSight; Fusion 360; working knowledge of SolidWorks; cross-functional technical training.

## PROFESSIONAL DEVELOPMENT

- Hands-on Siemens PLC development using S7-1200 and LOGO!: Ladder Logic, timers/counters, seal-in logic, interlocks, sequencing, basic HMI concepts and machine-troubleshooting exercises.
- Continuing development in industrial automation topics including VFD/servo/encoder fundamentals, industrial communications, barcode/downtime integration and motion-control concepts.

## EDUCATION

BCIT — Advanced Certificate in Industrial Automation and Mechatronics (ACIM) | Part-time, in progress | 2026–Present
**Bachelor of Technology - Electronics and Communication Engineering** | 2015-2018 | Punjabi University, Patiala, India
**Diploma in Technology - Electronics and Communication Engineering** | 2012-2015 | JSSPC, Patiala, India
**International Business Management** | St. Lawrence College, Cornwall, Ontario, Canada